

# THE FIRST AND SECOND MOMENT METHODS

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## GIVE ME A MOMENT

Moments have a long history in probability and statistics, with the method of moments famously being formalised by Karl Pearson around the beginning of the 20<sup>th</sup> century. Modern portfolio theory made the first two moments especially important in finance. The first two moments provide a simple summary of a potentially complicated probability distribution. Whilst mean and variance can be informative, they don't capture everything. For example, financial returns can exhibit fat tails, skewness, *etc.*, therefore, the first and second moment methods are helpful because they simplify the problem, but do note that simplification can lead to a loss of information.

The first moment of a random variable is its expected value,  $\mathbb{E}[X]$ . The second moment measures the expected squared size of  $X$ ,  $\mathbb{E}[X^2]$ . And in general, the  $k^{\text{th}}$  moment is  $\mathbb{E}[X^k]$ . This definition of moment is remarkably similar to the definition in physics, where a physical moment measures how mass or force is distributed across a distance relative to a reference point. However, instead of mass, we use probability mass, and instead of distance, we use the  $X$  axis of the distribution.

The first and second moment methods are applications of Markov's and Chebyshev's inequalities respectively. While we can apply Markov's inequality to any non-negative random variable and Chebyshev's inequality to any random variable, they are particularly powerful when applied to counting random variables, and we can use these results to see how to turn many interesting problems in probability into questions about counting random variables.

## MARKOV'S INEQUALITY AND THE FIRST MOMENT METHOD

Let  $X$  be a non-negative random variable, then Markov's inequality states that  $P(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$ .

Say we have a counting, or any non-negative, random variable  $X$  and its expectation is small. Then Markov's inequality tells us that  $X$  cannot be too big too often. Often we want to use the first moment method to show that typically, with probability very close to 1, no bad events occur. Thus leading to the first moment method for counting random variables. Let  $B_1, B_2, \dots, B_m$  be a sequence of bad events that may depend on some hidden parameter,  $n$ . If  $\lim_{n \rightarrow \infty} \sum_{i=1}^m P(B_i) = 0$ , then  $\lim_{n \rightarrow \infty} P(\text{At least one bad event occurs}) = 0$ .

## CHEBYSHEV'S INEQUALITY AND THE SECOND MOMENT METHOD

The first moment method tells us that if the expectation of a counting random variable is very small, then the probability the random variable is large is also very small. Let us ask the question, what if we have the opposite? Here, the expectation of  $X$  would be very large, therefore, can we say that  $X$  is likely to be large, or that the probability  $X$  is small is very small? Unfortunately, we cannot. Consider the following, let  $X_n = 0$  with probability  $1 - \frac{1}{n}$  and  $X_n = n^2$  with probability  $\frac{1}{n}$ . Then  $\mathbb{E}[X_n] = n \rightarrow \infty$  as  $n \rightarrow \infty$  but  $\lim_{n \rightarrow \infty} P(X_n = 0) = 1$ .

The expectation doesn't give us any useful information about the typical value of the random variable. In this case, the variance of  $X_n$  is  $\text{Var}(X_n) = \frac{n^4}{n} - n^2 = n^3 - n^2 \sim n^3$ , which is incredibly large. Informally, Chebyshev's Inequality expresses the opposite idea; that is, if the variance of a random variable  $X$  is not too large, then its expectation is a good guide to its typical behaviour.

Let  $X$  be any random variable with finite mean and variance, then Chebyshev's inequality states that  $P(|X - \mathbb{E}[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}$ .

Therefore, we can apply Chebyshev's inequality to counting random variables. Let  $X$  be a non-negative, integer-valued random variable, then  $P(X \geq 1) \geq 1 - \frac{\text{Var}(X)}{(\mathbb{E}[X])^2}$ . Or in other words, let  $B_1, B_2, \dots, B_m$  be a sequence of bad events that may depend on some hidden parameter,  $n$ . Let  $X = \sum_{i=1}^m \mathbf{1}_{B_i}$  be the number of these events that occur. If  $\lim_{n \rightarrow \infty} \frac{\text{Var}(X)}{(\mathbb{E}[X])^2} = 0$  then  $\lim_{n \rightarrow \infty} P(\text{At least one bad event occurs}) = 1$ .

Note that the use of the Paley-Zygmund inequality provides the famous second moment bound,  $P(X \geq 1) \geq \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}$ , since  $\frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]} = 1 - \frac{\text{Var}(X)}{(\mathbb{E}[X])^2 + \text{Var}(X)}$ , which is a slightly stronger version.

## AN EXAMPLE

Suppose  $X$  denotes the number of defaults in a portfolio, and  $\mathbb{E}[X] = 0.2$ ,  $\mathbb{E}[X^2] = 0.3$ . Use the first and second moment methods to bound the probability of observing at least one default.

*Solution:* The use of Markov's inequality yields  $P(X \geq 1) \leq \frac{\mathbb{E}[X]}{1} = \mathbb{E}[X] = 0.2$ . The use of the second moment method yields  $P(X \geq 1) \geq \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]} = \frac{(0.2)^2}{0.3} = \frac{2}{15}$ . Thus,  $\frac{2}{15} \leq P(X \geq 1) \leq 0.2$ . In credit risk, the first and second moment methods can be incredibly useful. Here, the first moment tells us how many defaults we expect, whilst the second moment tells us about how concentrated or dependent those defaults can be.

## CONCLUSION

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Although moments cannot fully describe the behaviour of a distribution, the first two moments often provide valuable information about risk. The first moment tells us how large the random variable is on average, while the second moment tells us whether that average is representative of its typical behaviour.

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